

Generative AI for Product Image Generation



Estimated time: 30 minutes

Welcome to the hands-on lab Generative AI for Product Image Generation.

Learning Objectives

After completing this lab, you will be able to:

- Demonstrate the use of generative AI to stimulate idea generation
- Demonstrate examples of image creation for marketing through generative AI

Scenario

Consider a well-established brand, 'Pure Glow', renowned for its high-quality skincare and cosmetics offerings. Recently, the brand has made the strategic decision to expand its product portfolio by launching a new soap named 'Harmony Wash'. The soap offers gentle and effective cleansing, hydration, and nourishment for the skin.

In this scenario, you will understand how Pure Glow can leverage the image generation capabilities of generative AI across various aspects of product development, such as creating design elements for a product's labeling and packaging or for social media campaigns.

Generative AI tool used in the project

The examples in this project are demonstrated using the generative AI classroom lab environment.

About generative AI classroom lab

The generative AI classroom allows you to write and compare your prompts to generate the desired text with real-time chat responses. Moreover, you can choose from multiple generative AI models and learn about their strengths and weaknesses.

You will have the generative AI classroom environment and instructions on one page in a browser. The instructions will be on the left half of the screen, and the generative AI classroom will be on the right half of the screen. You can interact with the language model using the message and chat fields.

The image shows a web application interface for a Generative AI lab. The left sidebar contains navigation icons and links. The main content area is titled "Generative AI for Product Image Generation" and includes a "Skills Network" logo, an estimated time of 30 minutes, learning objectives, a scenario description, and information about the generative AI tool used. On the right, a chat interface is visible, titled "untitled chat", with a model selection dropdown set to "meta-llama/llama-3-2-1b-instruct", a "PROMPT INSTRUCTIONS" section, and a "How to use AI Classroom" guide with four steps: 1. Name the chat, 2. Choose your model(s), 3. Set the prompt instructions, and 4. Chat with your language model. The chat interface also includes a "Start chat" button and a "powered by watsonx" logo.

Generative AI for Product Image Generation

Skills Network

Estimated time: 30 minutes

Welcome to the hands-on lab Generative AI for Product Image Generation.

Learning Objectives

After completing this lab, you will be able to:

- Demonstrate the use of generative AI to stimulate idea generation
- Demonstrate examples of image creation for marketing through generative AI

Scenario

Consider a well-established brand, 'Pure Glow', renowned for its high-quality skincare and cosmetics offerings. Recently, the brand has made the strategic decision to expand its product portfolio by launching a new soap named 'Harmony Wash'. The soap offers gentle and effective cleansing, hydration, and nourishment for the skin.

In this scenario, you will understand how Pure Glow can leverage the image generation capabilities of generative AI across various aspects of product development, such as creating design elements for a product's labeling and packaging or for social media campaigns.

Generative AI tool used in the project

The examples in this project are demonstrated using the generative AI classroom lab environment.

About generative AI classroom lab

The generative AI classroom allows you to write and compare your prompts to generate the desired text with real-time chat.

How to use AI Classroom

- 1 Name the chat
- 2 Choose your model(s). You may select multiple models when "Compare Models" toggle is active
- 3 Set the prompt instructions
- 4 Chat with your language model

Want to elevate your AI learning experience? Check out [IBM watsonx.ai](https://ibm.com/watsonx.ai) now!

Type your message Start chat

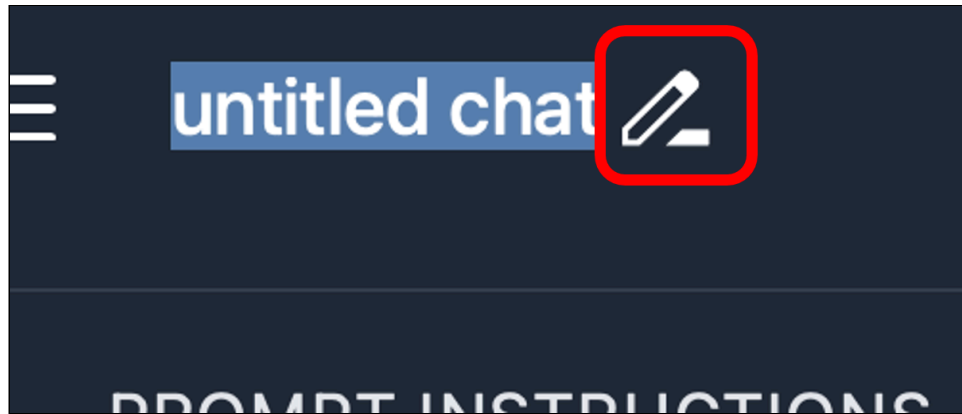
powered by **watsonx**

Exercise 1: Setting up the environment

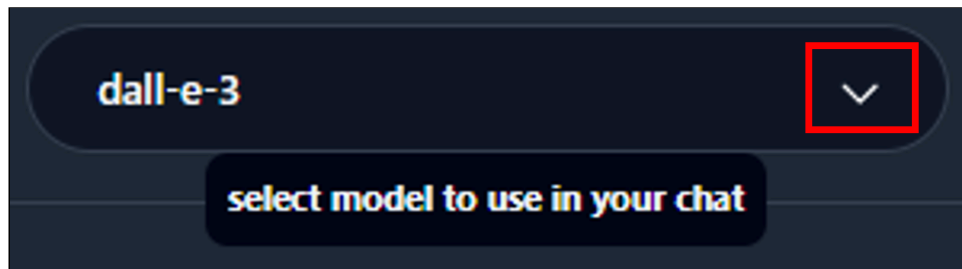
As a first step, you must set up your AI classroom for a better learning experience.

Before beginning the exercise, you must:

1. Name the chat: Use the pencil icon available on the top-left corner of the right pane to name the chat.



2. Choose the model: Use the dropdown option from the top-right corner of the right pane and select dall-e-3 for the image generation exercise.



3. Give prompt instructions: Use the **Prompt Instructions** field on the upper right pane of the chat system to provide instructions or any specific details about the context of the required output. These instructions will be locked when you start the chat and cannot be modified later.

PROMPT INSTRUCTIONS 

Write your prompt here. Instructions cannot be modified after sending your first message.

4. Type your message: Use the text box available at the bottom of the page to write the prompts and converse with the chat system.

Type your message

Start chat

Exercise 2: Generate the image for packaging design

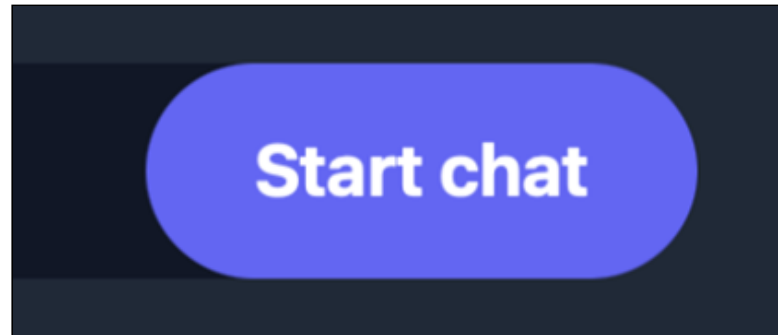
1. Consider the context of the image and draft the relevant prompt to generate the design for the soap bar packaging. The prompt should include the required details, such as the product name to be displayed on the packaging and any other element (such as the product's ingredients) on which you want the theme of the packaging to be based.
2. Let's enter the following prompt in the **Type your message** field

Generate a captivating and elegant packaging design for a body soap bar named Harmony Wash. Highlight the key ingredients like botanical extracts and essential oils.

Generate a captivating and elegant packaging design for a body soap bar named Harmony Wash. Highlight the key ingredients like botanical extracts and essential oils.

Start chat

2. Now, click **Start chat** to send the message to the chatbot.

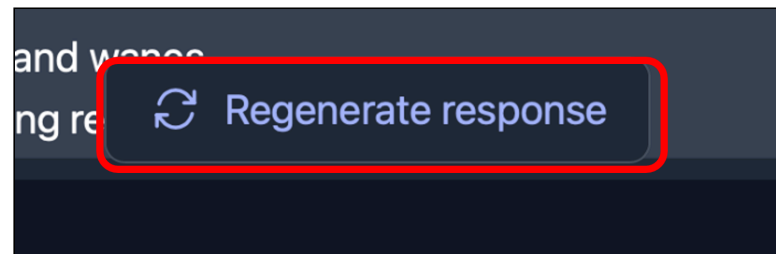


3. After a few seconds, the image will be generated. The image that was generated at our end is displayed here.





4. If you are unsatisfied with the response, click **Regenerate response**, and the chat system will show a revised response for your original prompt.

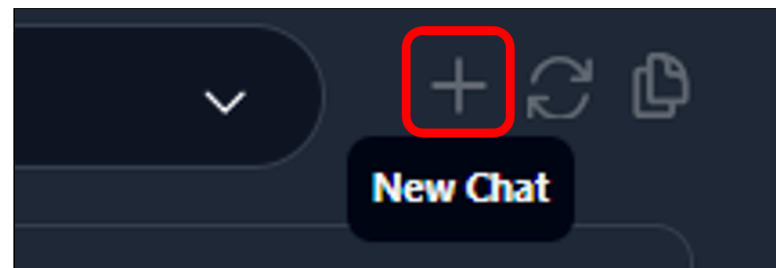


5. You can save the generated image by right-clicking on it and then selecting **Save image as**.

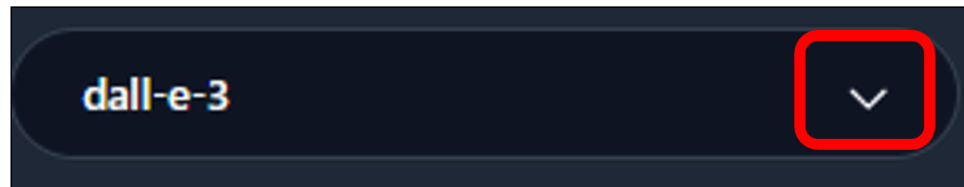
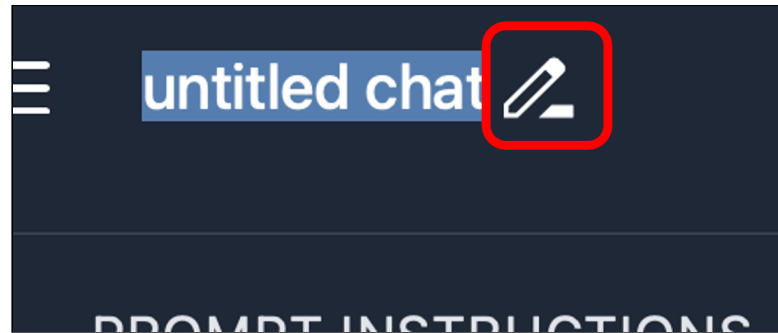
Exercise 3: Branding and marketing

Branding and marketing images can help with advertisements, social media posts, website banners, and promotional materials. Generative AI can assist in creating visually appealing graphics and layouts that align with your brand's aesthetic.

1. Select **New Chat** by clicking on the plus sign at the upper right corner of your classroom.

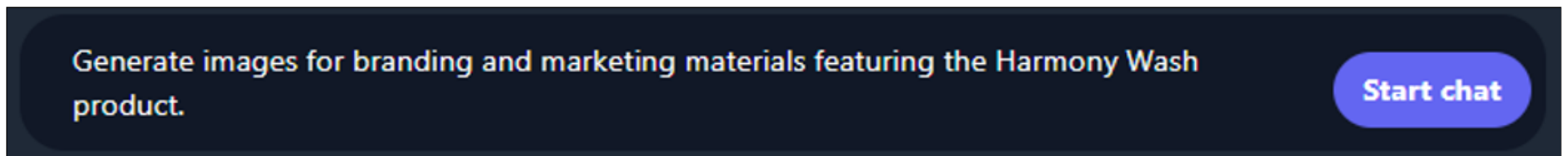


2. Name the chat and select the **dall-e-3** model.



2. Let's enter the following prompt in the **Type your message** field at the bottom of the screen.

Generate images for branding and marketing materials featuring the Harmony Wash product.



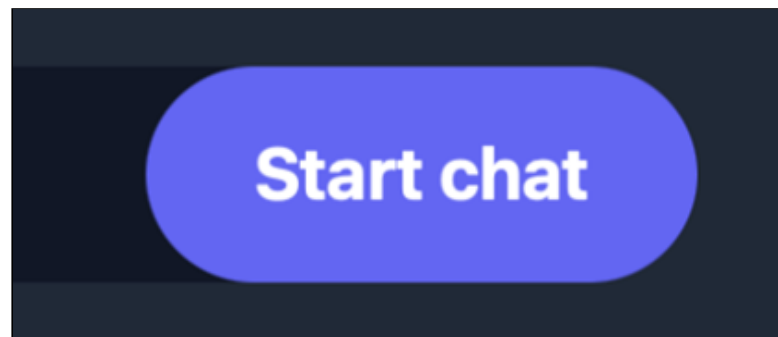
3. Include additional information or specific instructions about the context of the required output in the Prompt Instructions field.
For example,

Cinematic image

PROMPT INSTRUCTIONS 

Cinematic image

4. Click **Start chat** to send the message to the chatbot.

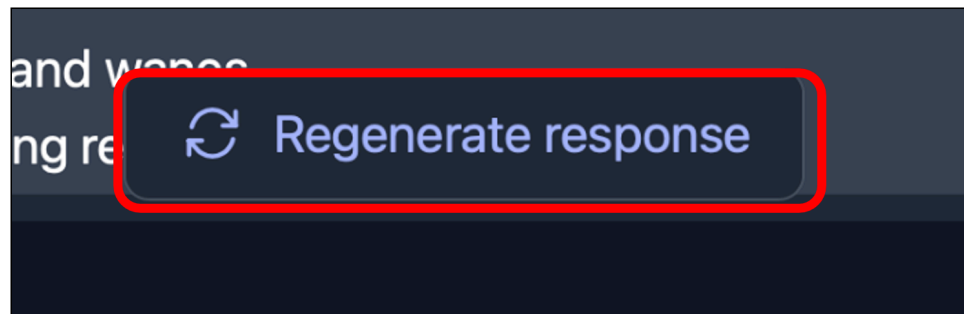


5. After a few seconds, the image will be generated. The image that was generated at our end is displayed here.

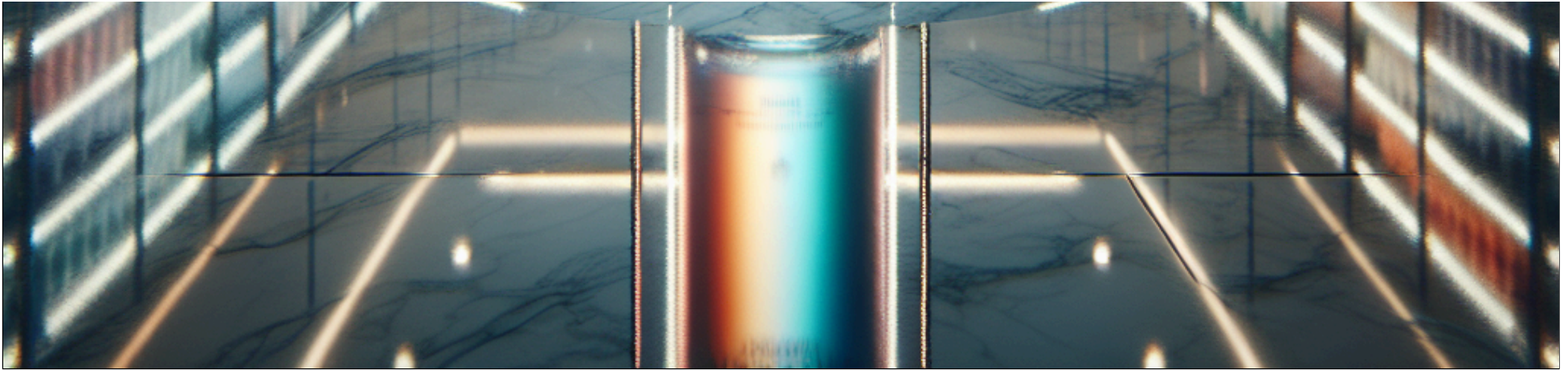




7. Let's see if we can get more images based on the same prompt by selecting **Regenerate response**. We get the following result:







8. You can save the generated image by right-clicking on it and then selecting **Save image as**.

So, generative AI can help you to produce multiple images based on a single prompt, giving you the flexibility to select the one that best aligns with your requirements and preferences.

Try yourself

1. Product demonstration

► [Click here for an example of a prompt](#)

2. Product ingredients

► [Click here for an example of a prompt](#)

Summary

Congratulations! You just completed the project Generative AI for Product Image Generation.

In this project, you explored a few use cases of image generation in generative AI related to the launch of a new product. You demonstrated generating images for packaging design, branding, and marketing.

Author(s)

Gagandeep Singh

© IBM Corporation. All rights reserved.